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PERFORMANCE OF INTRA- AND INTER-POPULATION SINGLE CROSS HYBRIDS OF MAIZE UNDER STRESS AND NON-STRESS ENVIRONMENTS

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The ultimate objective of an applied maize (*Zea mays* L.) breeding programme is to identify high yielding, environmentally stable maize varieties. Single-cross (SC) hybrids of maize are generally higher yielding than other types of hybrids or open pollinated varieties (Lynch *et al.*, 1973; Frey, 1983). Besides yield level *per se*, consistent performance of a hybrid across locations and over growing seasons (termed yield stab ty in space and time) is crucial to the maize farmer as it avoids the risk of loss associated with growing environmentally unstable varieties. A stable variety is one with one unit regression ($\beta = 1$), minimal standard deviation from regression and a high mean yield (Eberhart and Russell, 1966). Stability depends on the magnitude of genotype x environment interaction working against it.

Environmental instability is often caused by fluctuations in rainfall amounts over sites and seasons. Such fluctuations may be overcome in part by using environmentaly stable maize varieties. The objective of this study was, therefore, to identify stable, high yielding SC hybrids for commercial use in the Kenya Highlands.

MATERIALS AND METHODS

The evaluation was grown and analysed in a randomised complete block design (Snedecor and Cochran, 1967) using 28 SC genotypes with three replications at three locations in Kenya, namely the National Agricultural Research Centre (NARC) Kitale; Top-Farm (in Kitale area but in a different climatic environment from that of NARC); and Lanet in 1986 and 1988. The genotypes were generated from an eight-parent array of four local inbred lines (K1, K2, K3 and K4) derived from Kitale Station Maize and four exotic inbred lines (E5, E6, E7 and E8) extracted from Ecuador 573 in a diallel mating design without reciprocal crosses.

Plots consisted of three rows of nine hills each, which were overplanted and later thinned to 33 plants (two plants on the end hills and one in between). A standard spacing of 75 x 30 cm, inter-row x intra-row, was used, giving a plot area of 7.425 m^2 and a plant density of 44,444 plants/ha. Locations were stressed in one of the two years. Lanet experienced natural drought stress in 1986, but not in 1988. A droughty season was used to produce stress by deliberate late planting at NARC-Kitale in 1986 and at Top-Farm in 1988. Rainfall at each of these locations adequately meets the water requirements of maize throughout the growing season if the crop is planted at the recommended time.

Standard crop husbandry recommendations were followed in raising the crop. Diammonium phosphate (18:46:0) fertiliser was applied at a rate of 80 kg P_2O_5/ha at planting. The crop was kept

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weed-free by applying Atrazine (Primagram, 40% w:v) herbicide over the soil surface. After thinning, the plants were side-dressed using ammonium sulphate nitrate (26%) fertiliser at the rate of 100 kg/ha. The crop was treated with Dipterex (3.5% Granules) applied into the whorl of each plant to control stalk borer (*Busseola fusca* Underw.).

Data were recorded, on plot basis, on the following traits: stand count after thinning, number of days from planting to 50% pollen shed (mid-anthesis), ear height of 10 random competitive plants measured in centimetres, stand count at harvest time, number of stalk-lodged plants, and weight and moisture content of usable shelled grain. Plot yields were expressed as tons/ha at 12.5% moisture content.

Analysis of variance (ANOVA), combined over locations and years, was carried out as described by McIntosh (1983). Years and locations were considered as random effects and entries as fixed effects, in conformity with the scheme of Comstock and Robinson (1978).

Regression of trait means on environmental index followed procedures described by Eberhart and Russell (1966), using a linear model of the type:

$$Y_{ij} = \bar{Y}_{i.} + b_i(\bar{X}_{.j} - \bar{X}_{..}) + \delta_{ij}$$

where Y_{ij} is the yield of the ith genotype at the jth environment (location-year combination), $Y_{i.}$ is the Y-axis intercept for the ith genotype, $x_{.j}$ is the mean of all genotypes at the jth environment, β_i is the regression coefficient of the ith genotype, and δ_{ij} is the deviation from regression. Regression coefficients were tested at $P \leq 0.05$ on the basis of the null hypothesis: $\beta_i = 0$ using a T-test (for T with df = e-1):

$$T = (\beta_i - 0) / se_\beta$$

where se_β is the standard error of the regression coefficient of the ith genotype and e is the number of environments sampled.

Genotypic correlations between traits were computed using genotypic effects obtained from expected mean squares in combined analyses of variance of data pooled over locations and years.

RESULTS AND DISCUSSION

Genotypic means over locations and years are presented in Table I for grain yield and other agronomic traits. Of these genotypes, K_n x E_n SCs are experimental commercial-type hybrids. Three SCs, namely K1 x E8, K4 x E8 and K3 x E6, were outstanding, yielding 7.50, 7.46 and 7.56 tons/ha respectively. However, only K1 x E8 had stable yield performance, with a regression coefficient of 1.128 ± 0.078, compared to a relatively less stable SC, K3 x E6, with β = 0.878 ± 0.310 (Table II).

The stability of a SC hybrid is vested in the individual line buffering, due to heterozygosity at the loci concerned, compared to other hybrid types which have both heterozygosity and heterogeneity (Lynch *et al.*, 1973). Results from this study show that SC hybrid K1 x E8 had the greatest predictability of response (se_β = 0.078), implying a high capacity to adjust to environmental stress, based on the Eberhart-Russell (1966) criteria. The best SC hybrid (K1 x E8) was 2 days less to mid-anthesis and had 24.6% more closed-tip ears, but it was 5.0 cm taller and had 11.6% more stalk-lodged plants than K3 x E6. Single cross hybrid K4 x E8 was taller and later than the first two SCs and had comparable lodging incidence. The highest yielding local x local SC (K3 x K4, 6.41 tons/ha) had the highest number of days to mid-anthesis and the highest percentage of open-tip ears. In contrast, the highest yielding exotic x exotic SC (E5 x E7, 4.55 tons/ha) had the lowest incidence of open-tip ears. The observation implies that Ecuador 573 (exotic) inbred lines are a good source of genes conferring tight husk cover. Husk cover expression of the parental lines was expressed in their SC progenies in a manner that suggested a non-additive genetic effect for this trait, at least in one case. For example, SC K3 x K7 had high (55%) whilst K1 x K5 had low (1%) incidence of open-tip ears. Plant lodging may have occurred for reasons other than genotype, due to late harvest at some of the locations.

TABLE I - GENOTYPE MEANS FOR YIELD AND AGRONOMIC TRAITS OBTAINED FROM EVALUATION OF SINGLE CROSS HYBRIDS OF MAIZE IN SIX LOCATION-YEAR COMBINATIONS IN THE RIFT VALLEY PROVINCE OF KENYA IN 1986 AND 1988

	Parent of Cross								
	K2	K3	K4	E5	E6	E7	E8	Mean	LSD(0.05)
				Yield (tons/ha)					
									1.23
K1	5.04	5.47	5.60	5.91	7.13	7.26	7.50	6.27	
K2	—	5.89	6.05	6.33	6.47	6.35	6.85	6.14	
K3	—	—	6.41	6.73	7.56	7.22	7.35	6.66	
K4	—	—	—	6.86	6.75	7.15	7.46	6.61	
E5	—	—	—	—	3.69	4.55	4.26	5.48	
E6	—	—	—	—	—	2.54	2.88	5.29	
E7	—	—	—	—	—	—	3.29	5.48	
E8	—	—	—	—	—	—	—	5.66	
				Ear height (cm)					
									1.3
K1	173	179	186	190	199	186	201	188	
K2	—	184	200	201	206	197	204	195	
K3	—	—	194	203	196	196	207	194	
K4	—	—	—	215	210	219	219	206	
E5	—	—	—	—	194	190	200	199	
E6	—	—	—	—	—	180	188	196	
E7	—	—	—	—	—	—	191	194	
E8	—	—	—	—	—	—	—	201	
				Days to mid-anthesis (number)					
									0.5
K1	100	99	101	98	97	96	97	98	
K2	—	103	103	100	100	100	101	101	
K3	—	—	104	100	99	97	100	100	
K4	—	—	—	100	103	102	102	103	
E5	—	—	—	—	106	105	106	103	
E6	—	—	—	—	—	105	106	102	
E7	—	—	—	—	—	104	101	101	
E8	—	—	—	—	—	—	102	102	
				Open-tip ears (%)					
									0.9
K1	0.7	2.0	0.6	1.0	1.4	1.3	1.2	1.2	
K2	—	18.2	5.6	3.9	15.5	22.9	9.1	10.8	
K3	—	—	20.8	12.3	25.8	55.0	21.9	22.3	
K4	—	—	—	4.1	6.5	18.0	3.3	8.4	
E5	—	—	—	—	2.0	2.3	1.5	3.9	
E6	—	—	—	—	—	1.4	0.7	7.6	
E7	—	—	—	—	—	—	3.0	14.8	
E8	—	—	—	—	—	—	—	5.8	

TABLE II - ENVIRONMENTAL STABILITY PARAMETERS FOR YIELD (tons/ha) OF SINGLE-CROSS MAIZE HYBRIDS OBTAINED FROM SIX ENVIRONMENTS IN THE RIFT VALLEY PROVINCE OF KENYA IN 1986 AND 1988

Local inbred line (seed parent)	Exotic inbred line (male parent) E5	E6	E7	E8
	b+ s.e.$_b$			
K1	0.949 ± 0.111	0.849 ± 0.152	0.900 ± 0.151	1.128 ± 0.078
K2	0.819 ± 0.104	1.047 ± 0.230	1.178 ± 0.162	1.045 ± 0.182
K3	0.963 ± 0.147	0.878 ± 0.310	0.797 ± 0.093	0.763 ± 0.322
K4	1.359 ± 0.163	1.072 ± 0.181	1.167 ± 0.207	1.086 ± 0.205

TABLE III - YIELD (tons/ha) BY YEARS (TWO) AND LOCATIONS (THREE) OF 16 COMMERCIAL-TYPE SINGLE-CROSS HYBRIDS GROWN UNDER DROUGHT STRESS (ST) AND NON-STRESS (NS) ENVIRONMENTS IN THE RIFT VALLEY PROVINCE OF KENYA IN 1986 AND 1988

	1986 NARC ST	Top-F NS	Lanet ST	Mean	1988 NARC NS	Top-F ST	Lanet NS	Mean	Overall mean
				Averaged over exotic inbred lines					
K1 x E.	4.47	8.32	4.94	5.91	7.33	7.06	9.59	7.99	6.95
K2 x E.	4.26	8.08	4.40	5.58	6.74	5.95	9.58	7.42	6.50
K3 x E.	4.76	8.12	5.70	6.20	8.62	6.60	9.51	8.24	7.22
K4 x E.	4.67	8.71	4.61	6.00	6.36	7.03	10.94	8.11	7.06
				Averaged over local inbred lines					
K. x E5	3.82	7.65	4.38	5.28	6.84	6.85	9.47	7.72	6.50
K. x E6	4.73	8.55	5.25	6.18	7.72	5.84	9.78	7.78	6.98
K. x E7	4.66	8.84	4.82	6.11	7.11	6.75	9.79	7.88	6.99
K. x E8	4.96	8.19	5.18	6.11	7.37	7.46	10.58	8.47	7.29
Mean	4.54	8.31	4.94	5.92	7.26	6.66	9.91	7.94	6.93
$LSD_{0.05}$	1.34	2.51	1.41	1.03	2.60	2.86	1.65	1.36	0.84

TABLE IV - MEAN SQUARE TEST ON GENOTYPES FOR YIELD AND OTHER AGRONOMIC TRAITS OF SINGLE-CROSS HYBRIDS AT EACH OF SIX ENVIRONMENTS IN THE RIFT VALLEY PROVINCE OF KENYA

	Yield tons/ha	Plant stand (%)	Open-tip ears (%)	Ear height (cm)	Days mid-anthesis (no)
1986:					
NARC-Kitale (ST)	***	ns	***	***	***
Top Farm (NS)	***	ns	***	***	***
Lanet (ST)	***	ns	***	**	***
1988:					
NARC-Kitale (NS)	***	ns	***	ns	***
Top Farm (ST)	***	ns	***	**	***
Lanet (NS)	***	ns	***	***	***
Combined over locations and years	***	ns	***	***	***

*, **, *** - Significant at $P \leq 0.05$, 0.01, 0.001, respectively
ns - Not significant

TABLE V - COMBINED MEAN SQUARES FOR GRAIN YIELD AND OTHER AGRONOMIC TRAITS FOR 28 SINGLE-CROSS HYBRIDS GROWN AT THREE LOCATIONS IN THE RIFT VALLEY PROVINCE OF KENYA IN EACH OF TWO YEARS (1986 AND 1988)

Source of variation	df	Yield (tons/ha)	Plant stand (%)	Open-tip ears (%)	Ear height (cm)	No. days to mid-anthesis
Years (Y)	1	33,652.8^{ns}	18,439.0^{ns}	2,030.4^{ns}	27,916.7^{ns}	1,426.8^{ns}
Locations (L)	2	14,707.9**	10,656.3**	679.8**	106,554.9**	150.4*
L X Y	2	38,949.8**	7,477.1**	5,945.2**	106,554.9**	150.4*
Replications /L/Y	12	859.5**	432.3^{ns}	189.9^{ns}	1,818.9^{ns}	32.5^{ns}
Genotypes (G)	27	3,890.0**	224.9^{ns}	2,635 5**	2,430.6**	149.3**
G x Y	27	306.8^{ns}	137.5^{ns}	57.8^{ns}	192.4^{ns}	6.2^{ns}
G x L	54	202.8^{ns}	124.1^{ns}	101.5^{ns}	194.9^{ns}	7.2^{ns}
G x L x Y	54	363.2**	111.0^{ns}	254.1**	208.4^{ns}	5.8*
Pooled error	324	170.1^{ns}	102.2^{ns}	46.4^{ns}	204.9^{ns}	4.2^{ns}

*, **, *** - Significant at $P \leq 0.05$, 0.01, 0.001, respectively
ns - Not significant

TABLE VI - PHENOTYPIC CORRELATION COEFFICIENTS (ABOVE DIAGONAL) AND GENOTYPIC CORRELATION COEFFICIENTS (BELOW DIAGONAL) FOR PAIRS OF TRAITS OF SINGLE-CROSS HYBRIDS MEASURED IN SIX ENVIRONMENTS IN THE RIFT VALLEY PROVINCE OF KENYA

	Yield (tons/ha)	Lodged plants	Ear height	Days to mid-anthesis
Yield	—	-0.118**	0.127**	-0.483
Lodged plants	-0.451***	—	0.434***	-0.097*
Ear height	0.612***	-0.764***	—	-0.242***
Days to mid-anthesis	-0.451***	-0.332***	0.045	—

*, **, *** - Significant at $P \leq 0.05$, 0.01, 0.001, respectively
ns - Not significant

Yields at each of the six environments (location-year combinations) averaged over the males for each local inbred line seed parent and over the seed parents for each exotic inbred line are presented in Table III. Averaged over locations, drought stress reduced yield by 37%. There was a 20% yield loss due to stress at Top Farm, 40% at NARC-Kitale, and 50% at Lanet. Increasing sensitivity to drought for local source inbred lines, assuming drought as the major cause of yield loss, was in the order: K1 (30% loss), K3 (32%), K4 (33%) and K2 (37%). For the exotic source inbred lines, drought sensitivity increased in the order: E8 (19%), E6 (21%), E7 (23%) and E5 (23%). The greatest reduction in yield was observed for SC hybrids K2 x E7 (53%) and K2 x E8 (54%) in 1986. Single cross K1 x E7 did not exhibit any yield loss in drought-stress environments compared to non-stress environments (data not shown).

The significance of genotypic variation determined from analyses within and across locations and years is shown in Tables IV and V, respectively. There were very highly significant differences among genotypes at all locations in both years for yield, number of days to mid-anthesis and percent open-tip ears (Table IV). Significant genetic variation ($P \leq 0.01$) was observed for ear height, except at NARC-Kitale (Table V) in 1988. There was no genotype x year interaction for any of the traits. Genotype x location interaction was significant only for number of days to mid-anthesis. Genotype x year x location interaction was significant ($P \leq 0.05$) for yield, days to mid-anthesis and percent lodging. There were no significant differences between years for any trait. However, highly significant ($P \leq 0.01$) differences were found between locations for all traits, except days to mid-anthesis. Location x year interaction existed for all traits.

Correlation coefficients are presented in Table VI. All pairs of traits were significantly ($P \leq 0.05$) correlated phenotypically. However, there were no genotypic correlations between ear height and days to mid-anthesis. This observation implies that it may be possible to select for ear height independently of days to mid-anthesis in these source backgrounds. Selection for lower ear placement need not necessarily affect genotypic maturity, a factor important for high grain yield potential of maize varieties which are recommended for the cool highlands of Kenya. Yield was positively correlated with ear height (ρ_g = 0.612***), but negativey correlated with number of days to mid-anthesis (ρ_g = -0.451***).

Recurrent selection pressure in the populations from which the inbred lines were extracted seems to have favoured taller, but earlier maturing, genotypes that had higher yield potential than the average.

CONCLUSIONS

There is sufficient genetic variability among the SCs of inbred lines extracted from the two populations, Kitale Synthetic II and Ecuador 573, for yield stability in space and time to warrant a concerted breeding effort in the development of such SC hybrids for the Kenya Highlands. The 28 experimental SC hybrids evaluated under stress and non-stress environments demonstrated that it should be feasible to develop high yielding, environmentally stable SC hybrids. Systematic selection for stability of performance in an index that incorporates grain yield, percent standing plants and stability parameters (β_i and δ_{ij}) should be possible.

SUMMARY

In a study to identify high yielding, stable single cross (SC) maize hybrids for commercial use, SCs were formed from eight lines. Three were from Kitale Station Maize, one from cycle two (C2) of Kitale Synthetic II and four from Ecuador 573 (at C2, C3, C4 and C6). Analyses were performed on data obtained from the evaluation of 28 SC hybrids in three replications at three high altitude locations in Kenya in 1986 and 1988. Each location experienced drought stress in one year and no stress in the other. Overall, yield was reduced by 37% under moisture stress. There was no genotypic variation for percent plant stand. However, significant ($P \leq 0.05$) genotypic variation, free of interactions, was observed for ear height. Differences among the genotypes for grain yield, open-tip ears and days to mid-anthesis were masked by genotype x location x year interaction. Experimental hybrids K3 x E6 and K1 x E8 were the highest yielding commercial type SC hybrids at 7.56 and 7.50 tons/ha, respectively. Of these, K1 x E8 had 5.0 cm lower ear placement, 24.6% less open-tip ears, 11% more standing plants and 2 days less to mid-anthesis than K3 X E6, and was deemed superior. Besides, SC K1 x E8 exhibited the highest stability over environments ($\beta = 1.128 \pm 0.078$), compared to K3 x E6 ($\beta = 0.878 \pm 0.310$). The best local x local SC (K3 x K4) yielded 6.41 tons/ha and the best exotic x exotic SC (E3 x E7) yielded 4.55 tons/ha. These two SCs were as tall, or taller, and at least 5 days later in anthesis than either K1 x E8 or K3 x E6. Husk cover expression of the parental lines was reflected in their SCs in a manner suggestive of a non-additive effect for this trait. Grain yield was correlated positivey with ear height, but negatively with days to mid-anthesis. Lack of correlation between ear height and days to mid-anthesis implied that selection for the two traits may proceed independently. The results obtained herein demonstrate the feasibility of developing high yielding, environmentally stable, commercial-type SC hybrids. Efforts should be made to identify high yielding seed parents in order to reduce risks and costs in seed production.

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